

Parham Ghasemloo Gheidari

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[Google Scholar](#)

Research Profile

I build systems where large-scale human participation and AI solve hard computational biology problems together. My flagship project, **Borderlands Science (BLS)**, a citizen science mini-game embedded in Borderlands 3, has collected **200M+ puzzle solutions from 5M+ players worldwide**, producing novel insights in microbial phylogeny. My work sits at the intersection of **computational biology**, **ML**, **collective intelligence**, and **HCI**, with applications across sequence alignment, phylogenetics, and population-scale microbiome analysis.

Education

PhD in Computer Science, McGill University, Montréal, Canada 2020 – present
Waldispühl Group, Structural Bioinformatics & Citizen Science Lab CGPA: 3.8/4

- Courses: Advanced Comp. Biol. Methods & Research, NLP, Human Computation and Citizen Science, Applied Machine Learning, Probabilistic Graphical Models, Reinforcement Learning, Intelligent Software Systems

BSc in Computer Engineering, Sharif University of Technology, Tehran, Iran 2015 – 2020
Major: Software Engineering CGPA: 17.65/20

- Course Highlights: Game Theory, Algorithmic Game Theory, Information Theory, Data Structures & Algorithms, Computer Networks, Operating Systems, Engineering Probability & Statistics

Publications

- **AI-augmented citizen science alignments enable phylogeny-aware, clade-resolved analysis of the human gut microbiome at population scale**
Ghasemloo Gheidari, P., Chang, K.-H., Butyaev, A., . . . , Lladser, M. E., McDonald, D., Sarrazin-Gendron, R., Knight, R., Waldispühl, J.
iMeta, 2026 (accepted)
- **Playing the Data: Video Games as a Tool to Annotate and Train Models on Large Datasets.**
Ghasemloo Gheidari, P., Chang, K.-H., Sarrazin-Gendron, R., Mutalova, R., Butyaev, A., Szantner, A., Waldispühl, J.
MoFA Workshop @ ICML, 2025 [\[Full text\]](#)
- **Improving microbial phylogeny with citizen science within a mass-market video game**
Sarrazin-Gendron, R., Ghasemloo Gheidari, P., et al.
Nature Biotechnology, 2024 [\[Full text\]](#)
- **Learning the Game: Decoding the Differences Between Novice and Expert Players in a Citizen Science Game With Millions of Players.**
Cai, E., Sarrazin-Gendron, R., Mutalova, R., Ghasemloo Gheidari, P., Blanchette, M., et al.
Foundations of Digital Games (FDG), 2024
- **Player-Guided AI outperforms standard AI in Sequence Alignment Puzzles**
Mutalova, R., Sarrazin-Gendron, R., Ghasemloo Gheidari, P., Cai, E., et al.
Collective Intelligence, 2023 [\[Full text\]](#)
- **Playing the System: Can Puzzle Players Teach us How to Solve Hard Problems?**
Mutalova, R., Sarrazin-Gendron, R., Cai, E., Richard, G., Ghasemloo Gheidari, P., et al.
CHI '23 (ACM CHI Conference on Human Factors in Computing Systems), 2023

In preparation

- Multi-agent extension of BLS Automated, with agents specialized in different sequential contexts
- A generalizable metagenomic binning method enabled by phylogeny-aware genome embedding

Talks and Posters

- **Borderlands Science Automated: Large-scale Reverberations Through Crowd Augmentation:** Flash talk and poster, Canadian Bioinformatics and Human Computation Conference (CBHC), 2026
- **Playing the Data: Video Games as a Tool to Annotate and Train Models on Large Datasets:** Poster, MoFA Workshop @ ICML, 2025
- **Citizen Science Platform for Accelerating Metagenomic Studies in Urban Built Environment:** Poster, D2R

Initiative, McGill University, 2025

- **Borderlands Science: Bringing Together Millions of Players for Human Health:** Talk and poster, McGill AMR Symposium, 2024

Research Experience

Research Assistant, Waldispühl Group, McGill University

Sep 2020 – present

- Designed, analyzed, and implemented novel algorithms in BLS for multiple sequence alignment (MSA) using millions of human puzzle solutions
- Designed Crowd Augmentation: a framework for sustainable citizen science using behaviorally cloned AI agents deployed as participants
- Designed and implemented BLS Automated (BLSA), instantiating crowd augmentation in a large-scale pipeline integrating MSA, phylogeny, and adaptive Haar-like distances (AHL2) to uncover taxonomic signals at population scale, in collaboration with Rob Knight's lab (UC San Diego)
- Re-engineered AHL2, a novel beta-diversity method, cutting memory usage $200\times$ to enable population-scale analysis. Released as an open-source QIIME 2 plugin.
- Prototyping interpretable representation learning of biological processes (e.g., molecular docking) for human-in-the-loop annotation and decoding
- Designed 3+ citizen-science game prototypes targeting distinct computational biology problems

Research Intern, Simon Fraser University (Prof. Leonid Chindelevitch)

Jul 2018 – Dec 2019

- Improved deconvolution models for pathogen strain heterogeneity in mixed microbial samples

Technical Skills

Languages: Python, C/C++, Java, JavaScript; working knowledge of R, C#, MATLAB

ML & Modeling: PyTorch, TensorFlow, scikit-learn, behavioral cloning, imitation learning, representation learning

Scientific Computing: NumPy, Pandas, memory- and compute-constrained algorithm design, multi-core Linux server workflows

Bioinformatics: QIIME 2 plugin development, multiple sequence alignment, phylogenetic reconstruction, large-scale phylogeny-aware beta-diversity

Teaching and Mentoring

PhD - McGill University

- Instructor – COMP 370/570 “Introduction to / Fundamentals of Data Science”, Fall 2026 (supervised by Derek Ruths)
- Guest Lecturer – INF889L “Human Computing, Crowdsourcing and Video Games”, UQAM (Instructor: Roman Sarrazin-Gendron)
- Head TA – COMP 330 “Theory of Computation” (Fall 2024) and COMP 251 “Algorithms and Data Structures” (Winter 2024, Winter 2022)
- TA, 13+ semesters – COMP 204, 250, 251, 302, 330, 462/561, 598, spanning intro programming, algorithms, theory of computation, computational biology, and machine learning (2021–2026)
- Research mentor – 4 undergraduate students; three contributed to BLS Automated and one leads a follow-up project building on it

BSc - Sharif University of Technology

- Head TA (Mgmt.) & Project Designer – ce40254 “Data Structures and Algorithms” (Prof. Ghodsi, Dr. Sedighin)
- TA & Assignment Designer – Game Theory, Algorithmic Game Theory, Analysis & Design of Systems, Computer Networks, Engineering Probability & Statistics, and Programming in Python/Java (2017–2019)

Service, Honors and Awards

- **Outstanding Reviewer Award**, ACM HCOMP 2026 (paper track). Reviewer for RECOMB 2026 (external), ACM CHI, and Collective Intelligence / HCOMP 2026
- **Graduate Research Enhancement and Travel (GREAT) Award**, McGill School of Computer Science: travel support for ICML 2025, Vancouver. [\[Details\]](#)
- **43rd place nationally** among 180,000+ applicants, Iranian National University Entrance Exam (Konkoor), Engineering Division, 2015